

System1 classifier comparisons for customer-care transcripts

Open encoders · TypeSafe.ai JEV · GPT-5.6 Sol and Luna

99.93%

Best accuracy

GPT-5.6 Sol

99.90%

JEV calibrated

4 / 4,050 errors

99.16%

Best local encoder

ModernBERT trained

1,000 synthetic conversations · 27 attributes · 700 train / 150 validation / 150 locked test

Mark Austin

Independent evaluation · September 19, 2026

Executive findings

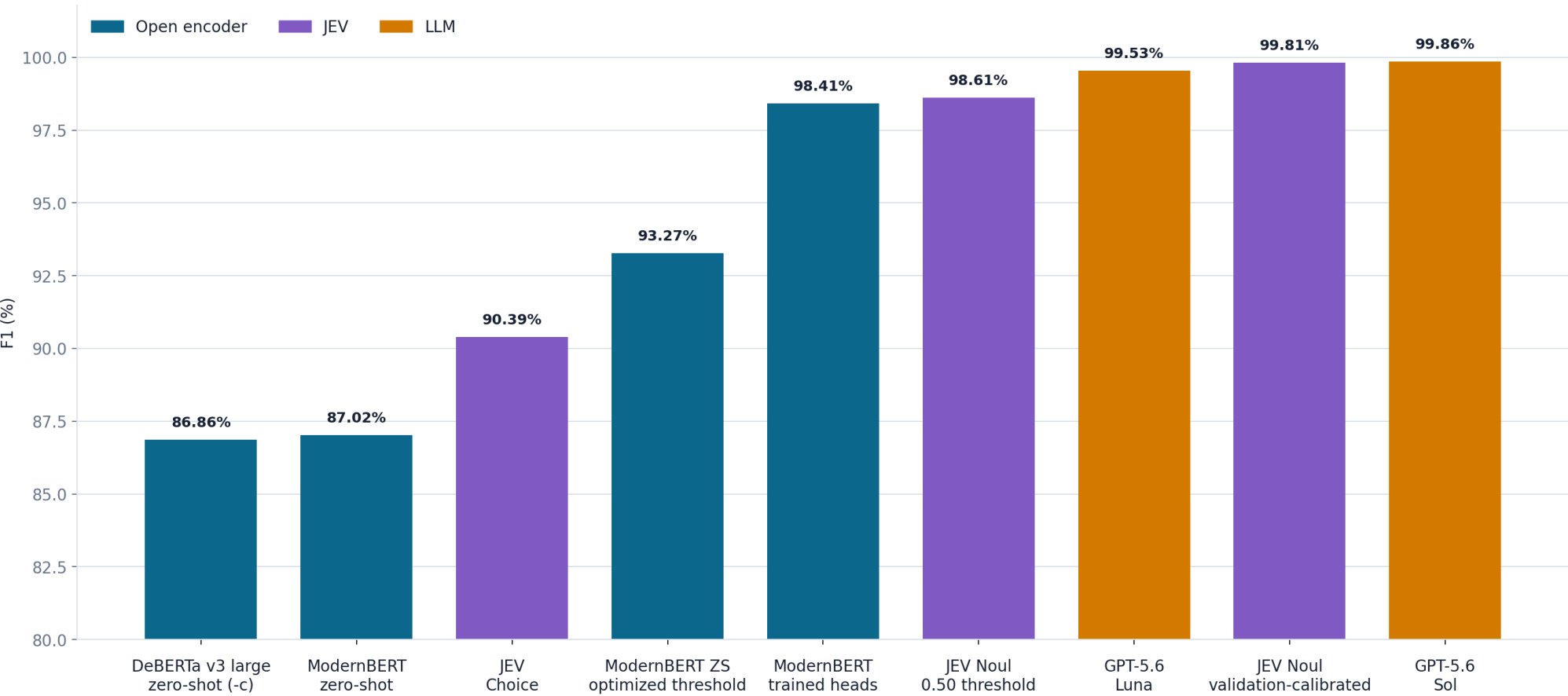
Quality, calibration, and operating-cost implications

- Sol achieved the highest measured quality: 99.93% accuracy and 99.86% micro F1, with three errors across 4,050 held-out decisions.
- Validation-calibrated JEV Noul was one decision behind Sol: 99.90% accuracy and 99.81% F1. Its estimated recurring API charge is about \$0.117 per 1,000 transcripts under the measured prompt shape.
- The strongest fully local result was frozen ModernBERT plus 27 supervised logistic heads: 99.16% accuracy and 98.41% F1.
- Threshold calibration alone improved unchanged zero-shot ModernBERT from 92.96% to 96.42% accuracy and cut errors from 285 to 145.
- DeBERTa-v3-large zero-shot (-c) did not beat ModernBERT zero-shot and was approximately 2.47x slower on the measured CPU path.
- For one transcript producing all 27 decisions, effective time was 293 ms for JEV Noul, 1,146 ms for trained ModernBERT, and 5,217 ms for zero-shot ModernBERT on the recorded paths.

Interpret carefully. This is a synthetic, template-driven smoke test. It is useful for controlled comparisons and pipeline validation, but it does not establish production accuracy. The JEV Choice and Noul runs also differ in state structure and criteria specificity, so their gap is not a primitive-only A/B result.

F1 comparison across transcript classifiers

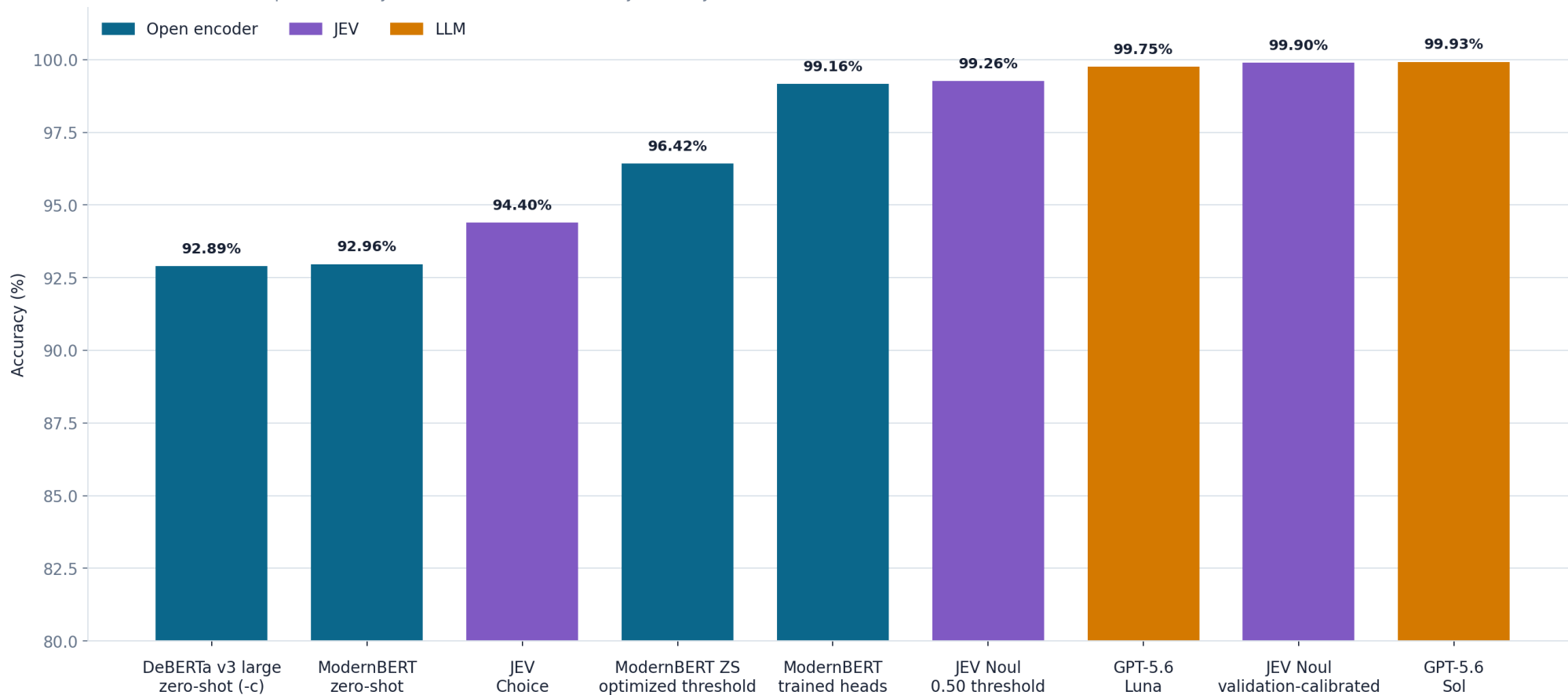
150 locked test transcripts · 27 binary attributes · models ordered by accuracy



F1 axis begins at 80% to make differences legible.

Accuracy comparison across transcript classifiers

150 locked test transcripts · 27 binary attributes · models ordered by accuracy



Accuracy axis begins at 80% to make differences legible.

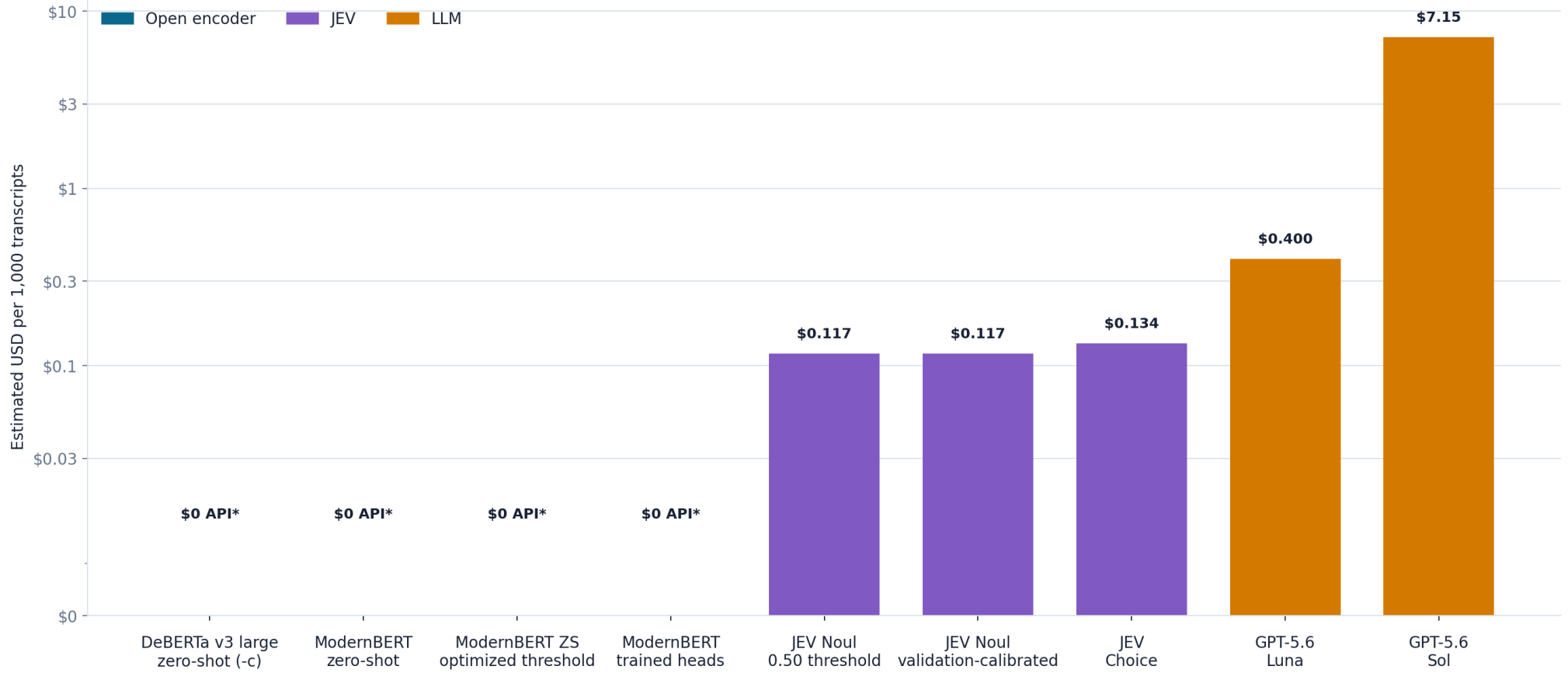
All transcript-classification quality metrics

Held-out test set · micro metrics across 4,050 decisions; exact match requires all 27 labels correct

Approach	Family	Accuracy	Precision	Recall	F1	Exact match
DeBERTa v3 large zero-shot (-c)	Open encoder	92.89%	84.70%	89.14%	86.86%	13.33%
ModernBERT zero-shot	Open encoder	92.96%	84.74%	89.42%	87.02%	14.00%
JEV Choice	JEV	94.40%	82.47%	100.00%	90.39%	18.00%
ModernBERT ZS optimized threshold	Open encoder	96.42%	92.53%	94.01%	93.27%	34.67%
ModernBERT trained heads	Open encoder	99.16%	98.05%	98.78%	98.41%	83.33%
JEV Noul 0.50 threshold	JEV	99.26%	97.27%	100.00%	98.61%	80.67%
GPT-5.6 Luna	LLM	99.75%	99.07%	100.00%	99.53%	93.33%
JEV Noul validation-calibrated	JEV	99.90%	99.63%	100.00%	99.81%	97.33%
GPT-5.6 Sol	LLM	99.93%	99.72%	100.00%	99.86%	98.00%

Estimated marginal inference cost for 1,000 transcripts

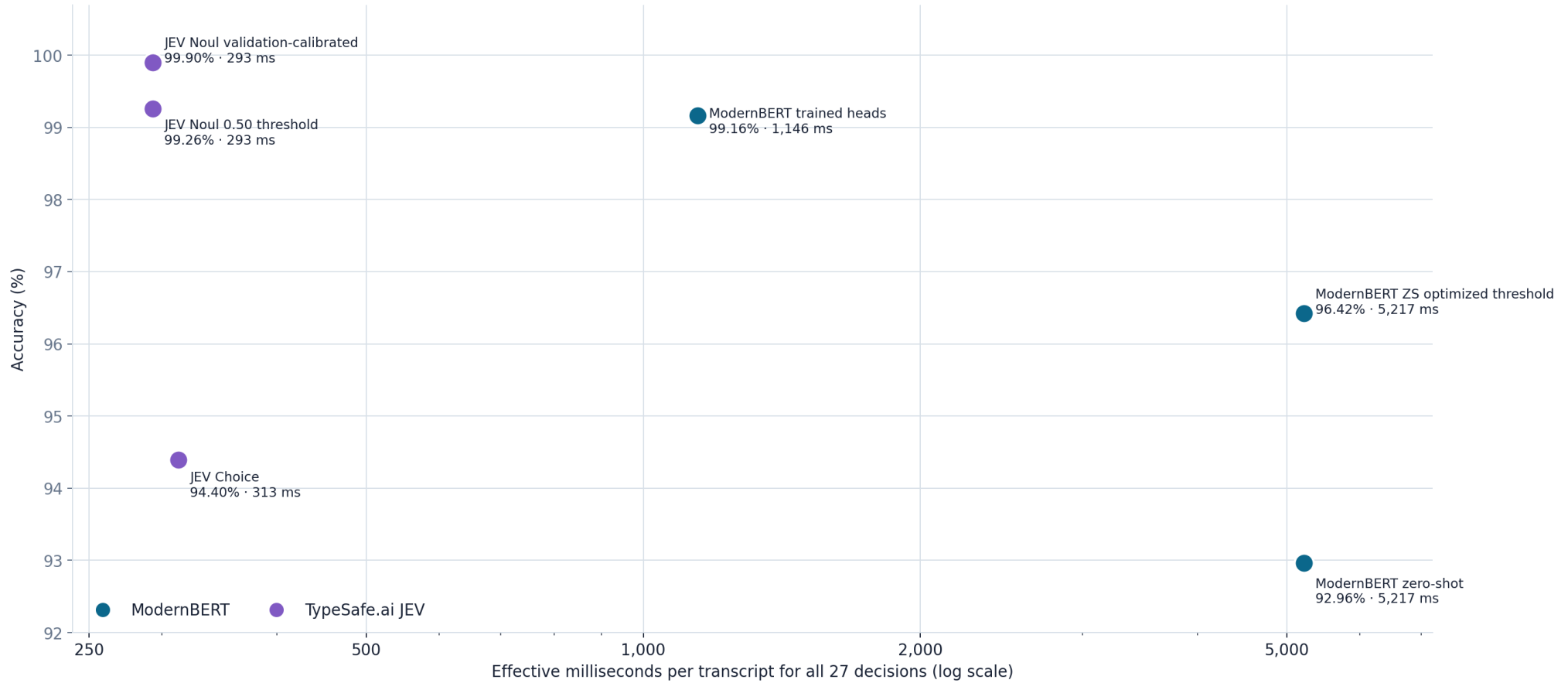
USD API charges · logarithmic scale above \$0.03 · validation/training and local hardware excluded



* Local encoders have no API fee; electricity, hardware, hosting, and operations are not zero and are not estimated. Sol/Luna use a standardized compact prompt estimate and exclude hidden reasoning tokens.

Accuracy at a common 27-decision transcript work unit

Effective wall-clock time per transcript · model load excluded · lower latency and higher accuracy are better



Same output unit, different deployment paths: ModernBERT is local CPU; JEV is hosted and includes network time. Values are elapsed/150 effective times, not batch-1 p50/p95 latency.

Comparable cost and latency units

Shared-state JEV accounting and one-transcript-to-27-decisions timing

JEV token accounting

Observed model: billable input $\approx$ state tokens + N x question tokens + fixed request overhead.

- The cost estimate does not multiply the full transcript state by all 27 questions.
- Observed latency stayed approximately flat from 1 to 16 questions, supporting parallel evaluation.
- The external behavior does not prove that the hidden neural architecture encodes state exactly once.

Equivalent latency work unit

- One transcript enters; all 27 binary decisions are available.
- JEV Noul: 293 ms effective time per transcript.
- JEV Choice: 313 ms effective time per transcript.
- Trained ModernBERT: 1,146 ms on the measured local CPU path.
- Zero-shot ModernBERT: 5,217 ms on the measured local CPU path.

Interpretation: the common work unit fixes the output obligation, not the hardware. JEV includes hosted network time; ModernBERT uses local CPU timing and recorded batching. Elapsed time divided by 150 is an effective throughput-derived time. A production SLA comparison still needs warm batch-1 p50/p95 on target hardware and matched concurrency.

Methodology and controls

What was held fixed and what changed

Dataset

- 1,000 synthetic alternating Agent/Caller conversations.
- 27 binary attributes per transcript.
- Scenario-family-stratified split: 700 / 150 / 150.
- Test truth excluded from training, prompt refinement, and threshold selection.

Metrics

- Micro accuracy, precision, recall, and F1 over 4,050 decisions.
- Exact match requires all 27 labels in a transcript to be correct.

Encoder variants

- Zero-shot: fixed premise/hypothesis NLI probabilities.
- Optimized threshold: 27 validation-selected scalar cutoffs; no weight training.
- Trained: frozen 1,024-d embedding plus 27 logistic heads and validation thresholds.

JEV and LLM variants

- JEV sends 27 typed questions in one request per transcript.
- Final Noul uses structured speakers and strict true/false boundaries.
- Sol and Luna produced blind binary JSON predictions before truth was joined.

Key distinction: trained ModernBERT learned approximately 27,675 logistic-head parameters from 700 rows. The optimized-threshold variant learned only 27 cutoffs from validation. The difference between their test results therefore estimates the value of supervised decision boundaries beyond operating-point calibration.

Conclusions, limitations, and sources

Practical interpretation

- For the highest measured quality, Sol led by one decision over calibrated JEV Noul.
- For low estimated API cost with near-frontier quality, calibrated JEV Noul was the standout in this synthetic task.
- For a fully local and controllable system, trained ModernBERT was strong, but it required labeled training data.
- Threshold calibration is worthwhile, but it does not replace supervised heads when enough labeled examples exist.

Limitations

- Synthetic templates are more explicit and regular than production conversations.
- Only 150 transcripts were held out; attribute decisions within a transcript are correlated.
- Choice versus Noul changed multiple factors and is not a clean primitive-only comparison.
- Latency paths and hardware differed; normalized times are throughput-derived, not matched-hardware batch-1 p50/p95 measurements.
- Sol/Luna standardized cost estimates exclude unavailable hidden reasoning-token usage.

Sources

- TypeSafe API: <https://docs.typesafe.ai/api>
- TypeSafe JEV launch/pricing: <https://typesafe.ai/blog/introducing-system-one-models-and-jev>
- OpenAI Sol pricing: <https://developers.openai.com/api/docs/models/gpt-5.6-sol>
- OpenAI Luna pricing: <https://openai.com/index/advancing-the-price-performance-frontier-with-gpt-5-6/>
- Model cards: <https://huggingface.co/MoritzLaurer>